

Seif Eddine Nefzi

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SUMMARY

Final-year Energy Engineering student motivated by the challenge of making energy systems cleaner, more efficient and more reliable. I turn ideas into working tools, from a microgrid energy management system that compares four optimization methods to a single-motor solar tracker designed and simulated for near dual-axis performance. Hands-on experience in energy modelling, system sizing, energy efficiency and industrial production analysis has taught me to move from data to decisions. I am seeking a 4 to 6 month graduation internship where I can contribute to real energy projects, learn from experienced engineers and keep growing.

EDUCATION

National Engineering School of Monastir (ENIM)	2024 – 2027
<i>Engineering Degree in Energy Engineering, Renewable Energies track</i>	<i>Final year</i>
University of Monastir (ENIM)	2025 – 2027
<i>Research Master's in Energy Systems Management</i>	<i>2nd year</i>
IPEIB, Bizerte	2021 – 2024
<i>Preparatory Classes, Physics-Chemistry</i>	

PROJECTS

EMS-MicroGrid, Hybrid Microgrid Energy Management System	2024 – Present
<i>IngENIM, ENIM Energy Club</i>	<i>Python</i>
Built an energy management system for a hybrid microgrid that compares four optimization methods: dynamic programming, genetic algorithm, particle swarm optimization (PSO) and linear programming, with a FastAPI backend.	
Single-Actuator Kinematic Solar Tracker	May 2026
<i>2026 IEEE Zucker Undergraduate Design Contest, individual submission</i>	<i>Supervisor: Prof. Z. Araoud</i>
- Designed a tracker that replaces the elevation motor with a passive ground rail derived from solar geometry, giving near dual-axis tracking with one motor. - Simulated annual performance in Python for Zarzis, Tunisia: cosine loss below 2% and a yield of 134% versus fixed tilt (simulation results). Estimated 60% lower CAPEX than a dual-axis tracker.	

EXPERIENCE

Photovoltaic Engineering Intern	June 2026
<i>Zenith Solar Engineering</i>	<i>Tunis</i>
Sized photovoltaic installations and carried out technical studies.	
Production Analysis Intern	July 2026
<i>CTKCP, Sidi El Kilani oil field</i>	<i>Tunisia</i>
Analyzed daily production reports (February to June 2026) and built monthly summaries and charts.	
Energy Efficiency Intern	August 2025
<i>SOTULUB, Tunisian Lubricants Company</i>	<i>Bizerte</i>
Analyzed the energy use of the used-oil regeneration process and studied waste heat recovery.	
Building Simulation and Sizing Intern	June 2025
<i>EcoBuilding Consulting</i>	<i>Tunis</i>
Performed energy simulation and sizing of buildings.	

LEADERSHIP AND ACTIVITIES

Head of Projects Committee	2024 – Present
<i>IngENIM, ENIM Energy Club</i>	<i>Monastir</i>
- Built SunBox Track, a dual-axis solar tracker controlled by Arduino/ESP32. - Lead prototyping of a mini wind turbine, a mini hydroelectric dam and an autonomous energy system.	
Active Member	2024 – Present
<i>IEEE ENIM Student Branch</i>	<i>Monastir</i>

CERTIFICATIONS

- Battery Energy Storage Systems for Grid Ancillary Services, RENAC (score: 96.67%)
- AI for Renewable Energy & Sector Coupling, RENAC Berlin (score: 100%)
- Introduction to Lighting and DIALux, Erasmus+ MELINA, ENIM (January 2026)

SKILLS

- Domains:** PV system sizing, hybrid energy systems, battery storage, microgrid energy management, optimization
- Software and tools:** Python, MATLAB, FastAPI, Arduino, ESP32, SolidWorks, AutoCAD (basic), DIALux, Excel, L^AT_EX
- Languages:** Arabic (native), French (fluent), English (fluent, technical)